

Divya Sreelekha Kara

AI & ML ENGINEER | COMPUTER VISION | DATA ANALYTICS | FULL-STACK AI APPLICATIONS

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PROFESSIONAL SUMMARY

Detail-oriented and analytical **AI & ML Engineer / CSE Undergraduate** (2027 Passout) certified in **Cisco Cybersecurity Essentials**. Experienced in designing automated software pipelines, building containerized applications, and deploying full-stack AI solutions. Proficient in **Python, Linux environments, and automation scripting** with strong fundamentals in Data Structures, OOP, and cloud-native concepts. Passion to leverage technical expertise in automation, cloud-native architectures (Docker, serverless), and security to support digital transformation and deliver robust software solutions for enterprise customers.

TECHNICAL SKILLS

Programming & Scripting	Python, JavaScript, C, SQL, Bash/Shell Scripting
Cloud & DevOps	Docker, AWS (basics), Modal Cloud (Serverless GPU), Linux CLI, Git/GitHub
Web & Databases	React.js, Node.js, Express.js, HTML5, CSS3, MongoDB, MySQL
ML/AI Frameworks	PyTorch, TensorFlow, OpenCV, MediaPipe, NumPy, Pandas, Scikit-learn, FastAPI
Security & Practices	Secure Coding, OWASP Awareness, Cybersecurity Fundamentals

EDUCATION

Vignan's Lara Institute of Technology & Science

B.Tech in Computer Science and Engineering (AI & ML)

2023 – 2027

CGPA: 8.7 / 10.0

- Key Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems (Linux/Unix), Database Management Systems, Computer Networks.
- Active participant in campus hackathons and technical seminars; served as team lead on cross-functional software teams.

TECHNICAL PROJECTS

C-THRU — Generative AI-Based Cloud Removal & Surface Reconstruction

2026 – Present

Python, PyTorch, cGAN, Streamlit, Rasterio, Linux

- Developing an operational deep learning framework (cGAN & DDPM Diffusion) to remove persistent cloud cover and shadow contamination from high-resolution **LISS-IV optical imagery**.
- Integrating cloud-penetrating **Sentinel-1 SAR** satellite imagery for topographical feature extraction, preserving terrain details under clouds.
- Built an interactive Streamlit dashboard for resample operations, coordinate alignment, and reconstructed GeoTIFF export (evaluating PSNR, SSIM, SAM).

RadReport — AI Chest X-ray Report Generator

2026

Python, FastAPI, OpenCV, React.js

- Built a clinical vision-language model (VLM) platform with a **FastAPI** backend that auto-generates structured radiology reports from uploaded chest X-rays, reducing manual reporting time.
- Implemented a dual-mode AI engine utilizing VLM inference with rule-based clinical heuristics fallback and **OpenCV** attention heatmap overlays for explainability.
- Delivered a full-stack deployment with a **React.js** frontend, enabling clinician-friendly report review, annotation, and PDF export workflows.

Nemotron Solver — Agentic Reasoning Model Fine-Tuning

2026

Python, PyTorch, LoRA, Jupyter

- Designed data augmentation pipelines and training corpus generation scripts; applied **LoRA** adapters for parameter-efficient supervised fine-tuning (SFT) on the Nemotron reasoning model (GSM8K-style datasets).
- Built serverless deployment and GPU orchestration workflows using **Modal Cloud**, automating model adapter updates.
- Designed a web-based local diagnostics dashboard to trace token-level generation traces, logprobs, and step-level metrics.

HANDY — Real-Time Hand Gesture Recognition System

2025

Python, OpenCV, MediaPipe

- Engineered a real-time hand gesture recognition system achieving **99% classification accuracy** at 20+ FPS by tracking 21 hand landmarks per frame using **MediaPipe** and **OpenCV**.

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WORK EXPERIENCE / INTERNSHIP

Machine Learning Intern

SkillCraft Technology

Mar 2026 – Apr 2026

- Implemented end-to-end supervised learning models (Linear Regression, SVM) and unsupervised models (K-Means Clustering) using **Python**, **Scikit-learn**, and **Pandas** on Kaggle Notebooks.
- Developed a hand gesture recognition pipeline using **OpenCV** and **MediaPipe**, achieving 99% accuracy through careful landmark feature engineering and classifier tuning.
- Conducted data preprocessing, exploratory data analysis (EDA), feature engineering, cross-validation, and performance benchmarking across assigned ML tasks.

CERTIFICATIONS & ACHIEVEMENTS

- ✓ **Cisco Cybersecurity Essentials** - Cisco Networking Academy
 - ✓ **HTML, JavaScript & Bootstrap** - Udemy
 - **Open Source Contribution:** Active GitHub contributor with a portfolio of ML and full-stack AI projects at github.com/KDSL1.
- ✓ **Python Fundamentals** - Infosys Springboard
 - ✓ **Sawit.ai Challenge** - GUVI (AI Problem Solving)